The Spike

Neural signalling III: the action potential

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Introduction to Neuroscience · Session 9 · Reading: Bear et al., Ch. 4; Rieke et al., *Spikes*, Ch. 1

Plan for today

Part	What we do	Time
Opening	One language for every sense; open Spike Lab	12 min
1 The spike	Threshold and all-or-none (Lab Part 1)	18 min
2 The doors	Sodium up, potassium down (Lab Part 2)	22 min
3 The pause	The refractory period (Lab Part 3)	13 min
4 The code	Adrian's discoveries, rate coding (Lab Part 4)	17 min
Wrap-up	Today on one page, again; exit ticket	8 min

You need: a laptop or phone (or a partner with one), the Session 9 booklet, a pen. **Lost?** Booklet page 2 has today on one page.

How these slides work

The bold line at the top of each slide is the one thing to remember.

Labels at the top right:

IDEA one idea **STUDY** a real finding

THINK a question for you

SPIKE LAB do this part of the lab

ANSWERS answers

Colours stay the same as Session 7:

Na^+ sodium **K^+** potassium

You can write instead of speaking, or pass. You do not need to copy slides.

Opening

One language for every sense



you are here

Warm-up

Think (writing or passing is fine)

A sunset, a song, the smell of coffee.

What actually travels from your eyes, ears and nose to your brain? Is it the same for all three, or different? Write a guess on booklet page 1.

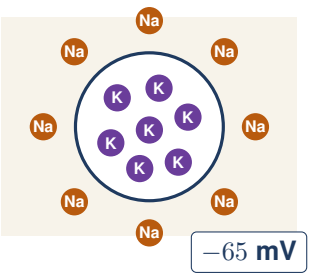
Every sense speaks the same language

Everything you sense reaches the brain as identical pulses called spikes.

- You do not see light. Your brain receives spikes from about **a million** optic nerve fibres.
- You do not hear sound waves. It receives spikes from about **thirty thousand** auditory nerve fibres.
- In the words of a classic book on the neural code: “Spike sequences are the language for which the brain is listening” (Rieke et al., 1997, Ch. 1).

Two things to remember from Session 7

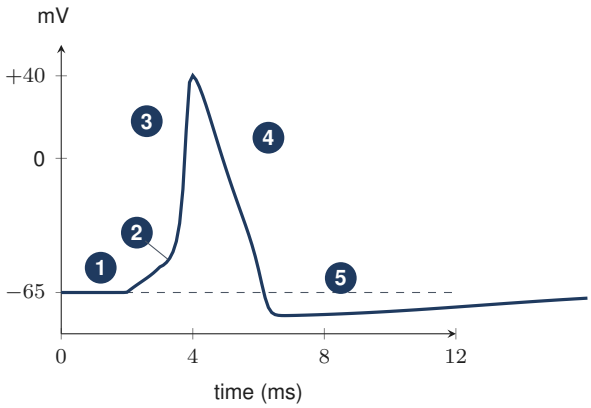
At rest the inside is -65 mV, and sodium is waiting outside to rush in.



- **K**$^+$ is crowded inside; **Na**$^+$ is crowded outside.
- Both pushes drive **Na**$^+$ **in**: it is crowded outside, and the negative inside pulls it.
- So if sodium doors open, **Na**$^+$ rushes in. That is the spike.

Today on one page

One spike, five moments. We take them one at a time.



1. **Rest** at -65 mV.
2. A push reaches **threshold**.
3. **Up**: Na^+ doors open, Na^+ rushes in.
4. **Down**: K^+ doors open, K^+ flows out.
5. **Dip and pause**: below rest; it cannot fire again yet.

Open Spike Lab now

Keep the lab open beside the slides. Each lab slide tells you which part.

- A model neuron you can **push** and watch. Works on a laptop or a phone.
- Four short parts, about 6–8 minutes each. Nobody sees your answers.
- **Nothing is saved or sent.** There are no sounds unless you switch them on.
- Press **Start Part 1**, then wait for the Part 1 slide.

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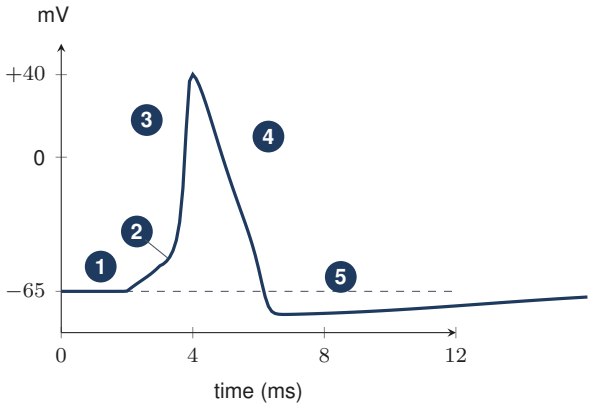
1 The spike

Threshold and all-or-none



What a spike looks like

Up in under a millisecond, down again, a small dip: about 2 ms in all.

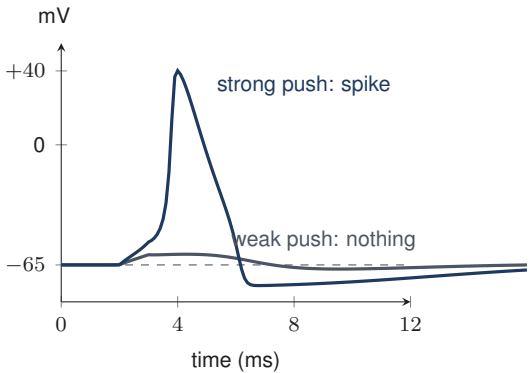


(Bear et al., Ch. 4)

- The inside flips from -65 to about $+40$ **mV**: briefly *positive*.
- It falls back, dips **below rest**, then recovers.
- Also called the **action potential**.

Threshold: all or none

Below threshold, nothing. Past threshold, a full-size spike, every time.



- Like flushing a toilet: a light press does nothing; past a point, you get the **whole** flush.
- A harder press does not give a *bigger* flush.
- The **threshold** is often around -55 mV, but it varies between neurons.

Spike Lab, Part 1: find the threshold

Time: 8 min | **Who:** alone or in pairs | **Where:** laptop or phone, booklet page 4

1. Set a strength and press **Stimulate**: one short push.
2. Find the **weakest** strength that makes a spike.
3. Read the peak at that strength, then at strength 10.
4. Type your numbers into the lab and press **Check**.

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labs/spike-lab

You are done when

The threshold and both peaks are in the booklet, and you answered the lab's question.

Spike Lab, Part 1: answers

Threshold at strength 4.5. The peaks are about the same: all or none.

Weakest strength that fires	**4.5** (at 4.0 the voltage only reaches about -59 mV)
Peak at threshold	about $+37$ **mV**
Peak at strength 10	about $+41$ **mV**

More than twice the push gave a spike only about 10% taller. The size of a spike tells the brain almost nothing about how strong the push was.

So far

A spike is a brief flip to positive, and it is all or none.

About 2 ms

From -65 mV up to $+40$ mV and back.

Threshold

A push must reach threshold.

All or none

Every spike is full size.

Next: what makes the voltage shoot up, and come back down?

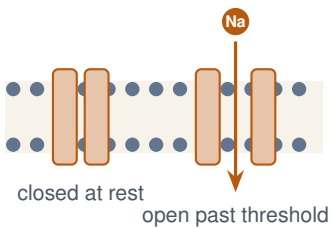
2 The doors

Sodium up, potassium down



Doors that respond to voltage

Some channels open when the inside becomes less negative.

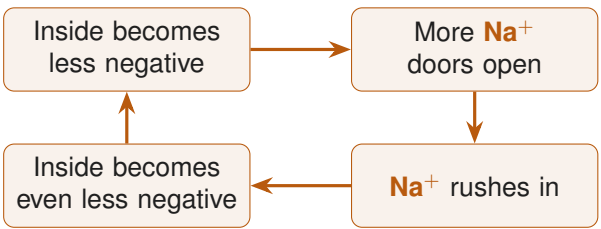


(Bear et al., Ch. 4)

- These are **voltage-gated channels**: a part of the protein senses the voltage and swings the door open.
- At rest they stay shut. Past threshold, many open together.
- There are Na^+ doors and K^+ doors, with different timing.

The chain reaction on the way up

Sodium in opens more sodium doors: the spike feeds itself.

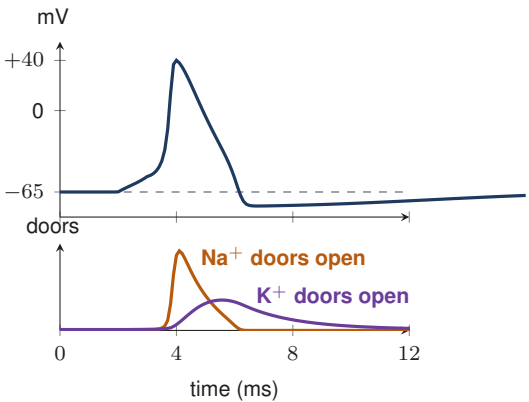


a loop that speeds itself up, like an avalanche

This is why there is a threshold: below it, the loop fizzles out; past it, nothing can stop it.

Two doors, two timings

Na^+ doors open fast, then shut themselves. K^+ doors open later and bring it down.



(Bear et al., Ch. 4; traces from the Hodgkin–Huxley model)

- Na^+ doors open within a fraction of a millisecond, then **lock shut** by themselves.
- K^+ doors open about a millisecond later. K^+ flows **out**, and the inside turns negative again.
- The K^+ doors close slowly, so the voltage dips **below rest**.

Squid, equations, and a single door

Hodgkin and Huxley explained the spike with a few equations, from a squid.

1. Using the squid's giant axon (the interlude story), Hodgkin and Huxley measured the sodium and potassium currents separately.
2. In 1952, their equations predicted a pulse whose shape and speed were essentially those of a real spike. Nobel Prize, 1963 (Rieke et al., 1997, Ch. 1).
3. In 1976, Erwin Neher and Bert Sakmann recorded the current through **one** channel opening and closing. Nobel Prize, 1991.

Spike Lab runs Hodgkin and Huxley's 1952 equations. You are using a model from 1952 that still works.

Spike Lab, Part 2: watch the doors, then block them

Time: 8 min | **Who:** alone or in pairs | **Where:** laptop or phone, booklet page 5

1. Press **Stimulate**. In the lower panel: which doors open first?
2. Tick the **sodium blocker** (like puffer-fish TTX) and stimulate.
3. Untick it; tick the **potassium blocker** (like TEA) and stimulate.
4. Note the time above 0 mV, with and without the potassium blocker.

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labs/spike-lab

You are done when

You answered the three questions and wrote the two times in the booklet.

Spike Lab, Part 2: answers

Na^+ brings the voltage up. K^+ brings it down.

Which doors open first?	Na^+ doors, then K^+ doors
Sodium blocker (TTX)	**no spike**: nothing can rush in
Potassium blocker (TEA)	the spike **lasts longer**: about 2.6 ms above 0 mV instead of 1.2 ms

Blocking a door, and seeing what breaks, is how scientists worked out what each door does.

So far

Two kinds of door, opening at different times, make the spike.

Na^+ in: up

Fast doors; a chain reaction past threshold.

K^+ out: down

Slower doors bring the voltage back.

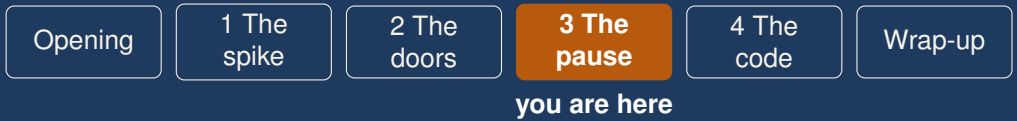
Block a door

No sodium: no spike. Less potassium: a longer spike.

Next: can a neuron fire again straight away?

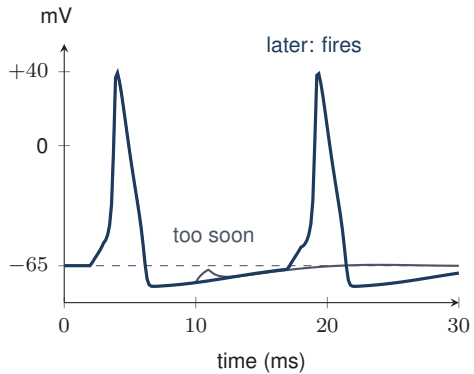
3 The pause

The refractory period



A pause after every spike

Right after a spike, the neuron cannot fire, then can fire only with a bigger push.



(Bear et al., Ch. 4; traces from the Hodgkin–Huxley model)

- **Absolute** refractory period: no push works; the Na^+ doors are locked. About **1 ms** in a warm mammalian neuron.
- **Relative** refractory period: a stronger push is needed.

Why the pause matters

The pause keeps spikes separate, one-way, and caps how fast a neuron can fire.

- **Separate:** each spike is a distinct event, like a click, not a hum.
- **One-way:** the patch just behind a travelling spike cannot fire, so the spike cannot turn back (Session 10).
- **A speed limit:** with about 1 ms of absolute pause, a neuron can fire at most about 1000 times a second (Bear et al., Ch. 4).

Spike Lab, Part 3: can it fire again straight away?

Time: 6 min | **Who:** alone or in pairs | **Where:** laptop or phone, booklet page 6

1. Two pushes, with a gap. Start at strength 8 and a short gap.
2. Lengthen the gap one step at a time until **both** pushes fire.
3. Repeat at strength 10. Does a shorter gap now work?

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You are done when

Both shortest gaps are in the booklet, and you answered the two questions.

Spike Lab, Part 3: answers

A stronger push fires sooner: the relative refractory period.

Shortest gap, strength 8	**13 ms**
Shortest gap, strength 10	**12 ms**

- The stronger push works sooner: some Na^+ doors are ready, but more push is needed.
- Very soon after a spike, no push works: the doors are locked.
- The model is a squid axon in cold sea water, so its pauses are much longer than your neurons' (about 1 ms).

So far

Every spike is followed by a short pause.

Absolute

No push works: Na^+ doors locked.

Relative

A bigger push is needed.

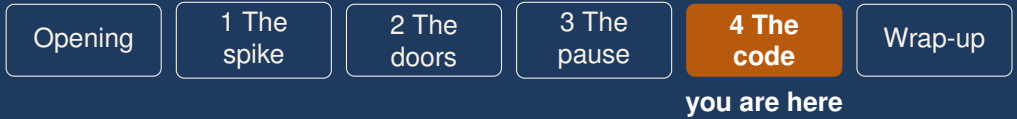
So

Separate, one-way spikes, with a speed limit.

Next: if every spike is the same, how does a spike train carry a message?

4 The code

Adrian's discoveries and rate coding



Why spikes at all?

A passive signal fades within about a millimetre. A spike renews itself as it travels.

- Without spikes, a voltage change spreads and fades over about **1 mm** (Hodgkin & Rushton, 1946, in Rieke et al., 1997).
- Your fingertip is about **1 metre** from your spinal cord.
- A spike is rebuilt at each point along the axon, so it arrives full size. The cost: energy, paid by the pump (Session 7).

Edgar Adrian's three discoveries (1926)

All-or-none spikes, a rate that tracks strength, and adaptation.

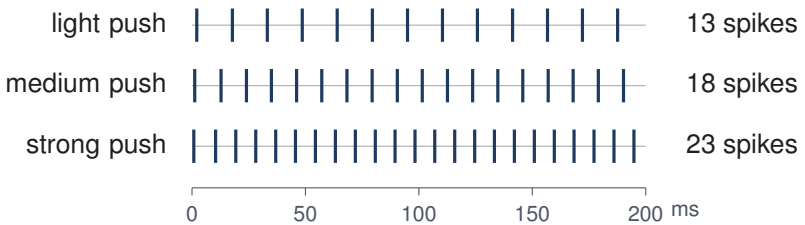
1. **All or none.** Single sensory neurons fire identical spikes.
2. **Rate coding.** Hang a heavier weight on a frog's muscle: its stretch receptor fires **more spikes per second**.
3. **Adaptation.** Keep the weight on, and the rate slowly falls.

(Adrian, 1926; Adrian & Zotterman, 1926; summarised in Rieke et al., 1997, Ch. 1)

Adrian linked adaptation to how we stop noticing a constant stimulus, like a watch on your wrist. Nobel Prize, 1932.

Stronger means faster

The message is in how many spikes arrive, not in their size.



Each tick is one spike in the Spike Lab model, during a 200 ms push. All the spikes are identical; only their number changes.

Spike Lab, Part 4: stronger means faster

Time: 6 min | **Who:** alone or in pairs | **Where:** laptop or phone, booklet page 7

1. The push now stays on for 200 ms.
2. Try strengths 1, 3, 5, 7 and 10.
3. Copy the spikes per second into the booklet table.
4. Optional: tick “hear spikes” to listen to them as clicks.

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labs/spike-lab

You are done when

Five dots are on the lab’s plot and the booklet table is filled in.

Stronger push, more spikes per second: rate coding.

Strength	Spikes per second
1	5 (one spike only)
3	80
5	90
7	105
10	115

- The rate rises with strength, just as Adrian found.
- The squid model jumps from one spike to about 65 per second. Real sensory neurons can also fire slowly.

Not just how many, but when

The exact timing of spikes can also carry information.

- A rate needs many spikes to average. But a fly's motion-sensing neuron often fires only about one spike per change in what it sees, and each spike can carry several **bits** (Rieke et al., 1997).
- A **bit** is the answer to one yes-or-no question.
- Give a cortical neuron the same flickering input twice, and it fires at nearly the same moments, to within about a millisecond (Mainen & Sejnowski, 1995).

Same spikes, different experiences

If all spikes are the same, why does seeing feel different from hearing?

- Johannes Müller (1830s): what you experience depends on **which nerve** is active, not on what activated it. This is the doctrine of specific nerve energies (in Rieke et al., 1997).
- Press gently on your closed eyelid: you may see light, though there is none.

Think (writing or passing is fine)

Is the "seeing" in the spikes, in where they go, or somewhere else? (Session 2: levels; Session 5: Leibniz's mill.)

Who reads the spikes?

Neuroscientists try to read spike trains, as if they were a little observer in the brain.

- The **homunculus** is a joke: a little person in your head watching the senses. It explains nothing, since who is in *its* head?
- Yet Rieke et al. argue that, when we record spikes and ask what they mean, we take the homunculus's place. Understanding the code means being able to do its job (Rieke et al., 1997, Ch. 1).

Think (writing or passing is fine)

If you could read every spike in a friend's brain, would you know what they experience?

So far

Identical spikes carry messages in their number and timing.

Why spikes

They travel a metre without fading.

Rate

Stronger stimulus, more spikes per second.

Timing

When spikes arrive can matter too.

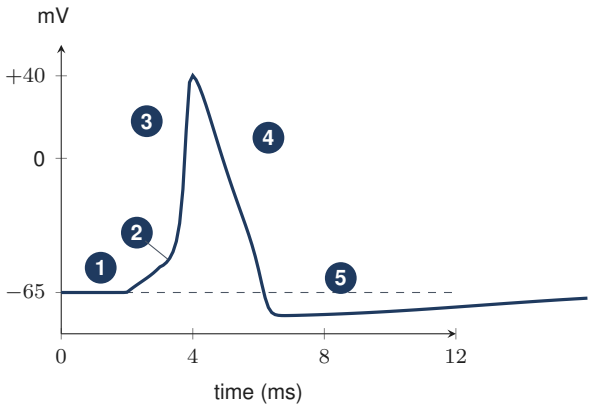
Wrap-up

Today on one page, exit ticket and next time



Today on one page, again

Can you explain each number to the person next to you?



1. **Rest** at -65 mV.
2. A push reaches **threshold**.
3. **Up**: Na^+ doors open, Na^+ rushes in.
4. **Down**: K^+ doors open, K^+ flows out.
5. **Dip and pause**: below rest; it cannot fire again yet.

Three things to take away

1. A spike is **all or none**: past threshold, Na^+ doors open in a chain reaction and the inside briefly turns positive.
2. K^+ doors bring it back down, and a short **refractory period** follows.
3. Spikes are identical, so the brain's messages are in **how many** spikes arrive and **when**.

Exit ticket and next time

Exit ticket: 3 minutes, booklet page 1

1. What does “all or none” mean?
2. Which ion brings the voltage up, and which brings it down?
3. How does a neuron signal a stronger stimulus?

Next session

How spikes **travel** along an axon, and why myelin makes them faster.

Read: Bear et al., Ch. 4.

Try: the Spike Lab free lab, at home.

Questions? rakesh.sengupta@krea.edu.in

References

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Adrian, E. D., & Zotterman, Y. (1926). The impulses produced by sensory nerve-endings. Part 2: The response of a single end-organ. *J Physiol* 61, 151–171.

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Rieke, F., Warland, D., de Ruyter van Steveninck, R., & Bialek, W. (1997). *Spikes: Exploring the Neural Code*. MIT Press. Chapter 1.

Spike Lab: Hodgkin–Huxley (1952) model, built with jsPsych 7 (de Leeuw, 2015, *Behav Res Methods* 47, 1–12). All plots in these slides come from the same model.

Check volume and page details before circulating.